

Sharp Newton-edge bounds and cancellation families with one nonzero quadratic norm

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Abstract

Let $g = (P, Q, R) = \sum_{i=0}^5 c_i x^{A_i}$ have six occupied vector coefficients, exactly one of which has nonzero norm for $\mathfrak{q}(P, Q, R) = PQ + R^2$. Over every field of characteristic different from two with a real-valued nonarchimedean valuation, we prove that $\mathfrak{q}(g)$ has at most eleven strict lower Newton edges and at most ten edges of width one. Both maxima are attained simultaneously over every such nontrivially valued field. At every prime, integer examples have coordinate sparsity fourteen and exactly ten simple nonzero local roots of distinct valuations. The dyadic example has 228-bit input coefficients. The upper bound is computer assisted: a generic null-cone perturbation and a rank-three initial-matrix argument reduce the problem to 2,359 marked support patterns. Exact disjunction trees and integer linear identities exclude every pattern, without an exponent or precision cutoff. Six patterns require convexity inequalities forced by equal barycenters of exponent sums. The supplied verifier uses only the Python standard library. We also prove the sharp five-support bound of eight edges and construct, at every prime and every support size $n \geq 8$, one-norm examples with $2n + 1$ edges, of which $2n$ have width one. The envelope lemmas and the family construction are proved in full.

1 The extremal problem

Let K be a field of characteristic different from two and let $\nu : K \rightarrow \mathbb{R} \cup \{+\infty\}$ be a nonarchimedean valuation. For $f = \sum_e f_e x^e \neq 0$, write $V(f)$ for the number of strict edges on the lower convex hull of the finite points $(e, \nu(f_e))$, and $U(f)$ for the number of those edges having horizontal width one. Collinear coefficient points do not create additional edges. Laurent exponents are allowed. Put

$$\mathfrak{q}(p, q, r) = pq + r^2, \quad g = (P, Q, R) = \sum_{i=0}^{n-1} c_i x^{A_i}, \quad A_0 < \cdots < A_{n-1}.$$

All c_i are nonzero. Let $k = |\{i : \mathfrak{q}(c_i) \neq 0\}|$, and let S be the sum of the three scalar support sizes.

Theorem 1.1. *If $n = 6$, $k = 1$, and $\mathfrak{q}(g) \neq 0$, then*

$$V(\mathfrak{q}(g)) \leq 11, \quad U(\mathfrak{q}(g)) \leq 10.$$

Over every K as above with nontrivial valuation, a formula attains $(V, U) = (11, 10)$. No bound on the input exponents or coefficient valuations is imposed. No completeness or residue-field assumption is needed.

Corollary 1.2. *For every prime p there are integer polynomials P, Q, R with $n = 6$, $k = 1$, $S = 14$, and $(V_p, U_p) = (11, 10)$. Their sole nonzero coefficient norm is an ordinary integer square. The output has exactly ten simple nonzero $\mathbb{Q}_p$ roots, of distinct valuations, and a separate double zero at the origin.*

The small-support results fit into a wider cancellation picture. Section 4 proves that five coefficient vectors with one nonzero norm have at most eight edges, and gives every-prime examples with eight unit edges. Together with Theorem 1.1, this makes six the smallest support size violating $V \leq 2n - 2$ over $\mathbb{Q}_p$. Section 8 constructs integer examples with $V_p = 2n + 1$, $U_p = 2n$, and $S = 3n - 4$ for every $n \geq 8$ and every prime. Their input bit lengths are $O(n^2 \log p)$. Thus the small-support extrema coexist with larger families exceeding $2n$ edges. All envelope arguments and constructions used for these statements are included below.

Sparse Newton-polygon bounds are motivated in part by algebraic complexity. Koiran, Portier, Tavenas, and Thomassé study a τ -conjecture for ordinary bivariate Newton polygons and show that a suitable weak version implies an arithmetic-circuit lower bound for the permanent [1]. Their polygon and representation model are different from the valued univariate, fixed ternary quadratic problem here. We use their work only as context; we do not infer its complexity conclusion from a six-support bound.

2 Reduction to finite pair and coefficient heights

Write $c_i = (P_i, Q_i, R_i)$ and set

$$d_i = \mathbf{q}(c_i), \quad C_{ij} = P_i Q_j + P_j Q_i + 2R_i R_j \quad (i < j).$$

The matrix M with diagonal $2d_i$ and off-diagonal C_{ij} has rank at most three, and

$$f = \mathbf{q}(g) = \sum_i d_i x^{2A_i} + \sum_{i < j} C_{ij} x^{A_i + A_j}.$$

Let a be the unique index with $d_a \neq 0$.

Lemma 2.1 (Generic null-cone perturbation). *After passing to a real-valued nonarchimedean field extension, the five null vectors can be perturbed so that every C_{ij} and every possible output coefficient is nonzero. The support and the number of nonzero coefficient norms remain unchanged, and every original strict Newton vertex can be preserved. Every original unit edge can also be preserved.*

Proof. Every nonzero null vector has the form $t_i(-u_i^2, v_i^2, u_i v_i)$ with $t_i \neq 0$. If $Q_i \neq 0$, take $t_i = Q_i$, $v_i = 1$, and $u_i = R_i/Q_i$. If $Q_i = 0$, nullity gives $R_i = 0$ and $P_i \neq 0$; use $t_i = -P_i$, $u_i = 1$, $v_i = 0$.

Replace u_i, v_i by $u_i + Z_i, v_i + W_i$, where the ten variables are algebraically independent and have a common sufficiently large positive Gauss valuation. Keep c_a fixed. The perturbed vectors are nonzero and null. Between two null vectors the cross coefficient is

$$-t_i t_j (u_i v_j - u_j v_i)^2,$$

which is a nonzero polynomial in the independent parameters. With c_a it is the nonzero quadratic form $t_i(-Q_a u_i^2 + P_a v_i^2 + 2R_a u_i v_i)$.

Pairs at a common exponent have disjoint endpoints. A null-null pair has a mixed monomial involving its own two endpoints, absent from all other contributions in that exponent group. A pair involving a has a nonconstant polynomial in its own null endpoint, likewise absent elsewhere in the group. Thus no possible output coefficient vanishes identically, even if a diagonal constant is present.

For each old strict vertex select a finite supporting slope exposing it uniquely. There are finitely many selected slopes and possible output exponents. Sufficiently large parameter valuations preserve all old nonzero coefficient heights and place every newly nonzero coefficient strictly above these selected supporting lines. All old vertices therefore persist. A unit edge persists because no integer exponent lies between its endpoints. $\square$

The use of a field extension is legitimate for the universal upper bound: the alleged counterexample would become another counterexample over a field in the same class. The lemma does not assert that a specialization can always be chosen inside the original field.

3 The hidden primary pair

For $t \in \mathbb{R}$ define the concave Gauss functions

$$\begin{aligned} E(t) &= \min\{\nu(d_i) + 2A_i t, \nu(C_{ij}) + (A_i + A_j)t\}, \\ D(t) &= \min_i \{\nu(d_i) + 2A_i t\}, \\ F(t) &= \min_e \{\nu(f_e) + et\}. \end{aligned}$$

Zero contributions are assigned infinite height. The following envelope lemma holds for arbitrary n and k ; its full proof supplies the preliminaries for both small-support bounds.

Lemma 3.1. *For rank at most three, $E = \min(F, D)$. If $E = F$, then $V(f) \leq 2n + k - 4$ for $n \geq 2$.*

Proof. Always $E \leq \min(F, D)$. If the inequality were strict on an interval, choose t there avoiding ties between distinct contribution exponents. Over $K(T)$ with the Gauss valuation $\nu(T) = t$, conjugate M by $\text{diag}(T^{A_i})$ and divide by a minimum off-diagonal entry times its monomial. All entries are integral, all diagonal reductions vanish, and the nonzero off-diagonal reductions are precisely the minimum pairs. There must be at least two such pairs, since the output coefficient is raised. Equal-sum pairs are disjoint, so the residue matrix contains two nonsingular two-by-two blocks and has rank at least four. This contradicts the vanishing of every four-by-four minor. Continuity gives the identity at exceptional parameters too.

Now assume $E = F$. At each strict output vertex, choose a contribution of that height. Such a contribution exists by the ultrametric inequality. At most k selected vertices use diagonals. Two strictly nested chosen off-diagonal pairs, on four ordered exponents $x_0 < x_1 < x_2 < x_3$, are impossible. Here is an explicit strictness argument. Write $s = x_0 + x_3$, $t = x_1 + x_2$, and let ϕ be the piecewise-linear Newton boundary, extended convexly with strict endpoint corners if needed. The selected reverse entries have heights $\phi(s), \phi(t)$, and both arguments are strict vertices. For sufficiently small $\eta > 0$, the function $\psi(x) = \phi(x) - \eta(|x - s| + |x - t|)$ is still convex. For any convex function, the reverse permutation minimizes the assignment sum $\sum_i \psi(x_i + x_{\pi(i)})$: successively swapping increasing permutation values cannot increase the sum, by the convex Monge inequality.

Every other permutation has an argument outside the interval with endpoints s, t . Indeed, $\pi(0) \neq 3$ gives an argument below both; $\pi(3) \neq 0$ gives one above both. The remaining nonreverse permutation fixes the middle indices, and one of $2x_1, 2x_2$ lies outside that interval. Therefore its sum of $|x - s| + |x - t|$ is strictly larger than the reverse sum. The reverse permutation uniquely minimizes the ϕ assignment cost. All matrix entries are at least their boundary heights, since $E = F$ (and $\nu(2) \geq 0$ for doubled diagonals), while the reverse entries attain their heights. Thus the reverse determinant term is uniquely minimum, contradicting rank at most three.

The selected pairs consequently form a nonnesting antichain. In lexicographic order both indices are nondecreasing and their sum strictly increases, through the range $1, \dots, 2n - 3$ in zero-based indexing. There are at most $2n - 3$ pairs. Adding k diagonal vertices gives $V + 1 \leq 2n - 3 + k$. $\square$

Corollary 3.2. *For $n \geq 2$, if every nonzero diagonal contribution lies on or above the output Newton boundary, then $V(f) \leq 2n + k - 4$. In particular, $V(f) \leq 2n - 4$ when all coefficient vectors*

are null. The bound $V(f) \leq 2n + k - 4$ also holds whenever the support contains no nontrivial three-term arithmetic progression.

Proof. The diagonal hypothesis says $D \geq F$, so Lemma 3.1 gives $E = F$ and the bound. For null vectors $D = +\infty$. If the support has no nontrivial three-term progression, no off-diagonal pair contributes at $2A_i$; hence $f_{2A_i} = d_i$ and every finite diagonal point belongs to the output coefficient cloud. $\square$

Lemma 3.3. *For $n = 6, k = 1$, any formula with $V > 9$ can be normalized to have a unique primary pair (p, q) satisfying*

$$\begin{aligned} p < a < q, \quad A_p + A_q = 2A_a, \\ w_{pq} = 0, \quad w_{ij} > 0 \ ((i, j) \neq (p, q)), \quad H_e > 0. \end{aligned}$$

Here w_{ij} are pair-entry heights and H_e are output heights after a common vertical translation and affine tilt. The normalized exceptional diagonal contribution has height zero. Reflection permits $a \in \{1, 2\}$.

Proof. Lemma 2.1 permits all pair and output coefficients to be nonzero. If $E = F$, Lemma 3.1 gives $V \leq 9$. Thus $E < F$ on an open interval. At a generic parameter in that interval, the unique diagonal at exponent $2A_a$ is minimal, and at least one pair at the same exponent must tie it. Such a pair avoids a and straddles it.

There cannot be two minimum pairs. They would give two disjoint nonsingular blocks in the residue matrix used above. With $\epsilon = \nu(2) > 0$ the diagonal vanishes on reduction; with $\epsilon = 0$ it adds a separate nonzero diagonal block. Either case contradicts rank at most three. Subtracting the minimum height and tilting by the chosen parameter gives the assertions. The extreme indices cannot be exceptional in a hidden primary, and reflection reduces $a = 3, 4$ to $a = 2, 1$. $\square$

4 The sharp five-support bound

Theorem 4.1. *If $n = 5, k = 1$, and $q(g) \neq 0$, then $V(q(g)) \leq 8$ over every field in the standing hypotheses. For every prime p , integer examples attain $V_p = U_p = 8$ with $S = 11$ and a sole nonzero coefficient norm that is an integer square.*

Proof of the upper bound. Suppose $n = 5, k = 1$, and, toward a contradiction, $V \geq 9$. Let a be the exceptional exponent. There are only eleven possible contributions: ten off-diagonal pairs and $d_a x^{2a}$. Lemma 3.1 implies $E \neq F$, and Lemma 3.1 then forces the diagonal to be hidden by cancellation. Counting contributions shows that all ten off-diagonal terms are nonzero with distinct exponent sums, precisely one pair $p < a < q$ satisfies $p + q = 2a$, and every one of the ten output points is a strict vertex. No output coefficient vanishes.

Put $d = \nu(d_a)$, $w_{ij} = \nu(C_{ij})$, and let h_{ij} be the actual output height at $i + j$. In this section the letters p, a, q, r, s denote actual support exponents, not their ordinal indices. Then

$$w_{pq} = d < h_{pq}, \quad w_{ij} = h_{ij} \quad (\{i, j\} \neq \{p, q\}). \quad (1)$$

Strict convexity gives, whenever $i < j < k < l$,

$$h_{ij} + h_{kl} > h_{ik} + h_{jl} > h_{il} + h_{jk}, \quad (2)$$

and, for $r \notin \{p, a, q\}$,

$$h_{pr} + h_{qr} > 2h_{ar}. \quad (3)$$

In (2) the paired arguments have equal totals and successively smaller spreads.

For four null vectors, the Gram determinant identity is

$$X^2 + Y^2 + Z^2 - 2XY - 2XZ - 2YZ = 0, \quad (4)$$

where X, Y, Z are the three matching products of cross coefficients. Their valuations cannot have a unique minimum. For the exceptional vector a and three null vectors p, q, r , the identity has the additional term

$$4d_a C_{pq} C_{pr} C_{qr}. \quad (5)$$

These are direct determinant expansions of M .

If another exponent r lies strictly between p and q , the nested matching $(p, q), (a, r)$ is uniquely smallest by (1)–(2), with valuation $d + w_{ar}$. The additional term has valuation $2\epsilon + 2d + w_{pr} + w_{qr} > 2d + 2w_{ar}$ by (3). The square of that matching product is the unique minimum determinant term, a contradiction.

Thus p, a, q are consecutive. If the other exponents lie on opposite sides, write $r < p < a < q < s$. On the four null vectors r, p, q, s the nested matching $(r, s), (p, q)$ is uniquely smallest, contradicting (4).

Otherwise reflect the exponents so that $p < a < q < r < s$. The four-null identity on p, q, r, s forces

$$d + w_{rs} = w_{ps} + w_{qr},$$

since the middle matching is strictly larger than the nested matching. On a, q, r, s the nested matching $(a, s), (q, r)$ is uniquely smallest. The additional determinant term must have exactly its doubled valuation:

$$2\epsilon + d + w_{qr} + w_{qs} + w_{rs} = 2w_{as} + 2w_{qr}.$$

Substitution gives $2\epsilon + w_{ps} + w_{qs} = 2w_{as}$, contradicting (3). This proves $V \leq 8$. $\square$

4.1 Sharpness with eight unit edges at every prime

Set $c_3 = (0, 2, 1)$ and $c_i = t_i(-s_i^2, 1, s_i)$ for $i = 0, 2, 4, 6$. In the dyadic case take

$$(s_0, s_2, s_4, s_6) = (16, 80, 2, 10), \quad (t_0, t_2, t_6) = (64, 1/2, 1),$$

and define

$$t_4 = \frac{1 - t_0 t_6 (s_0 - s_6)^2 - 4}{t_2 (s_2 - s_4)^2} = -\frac{769}{1014}.$$

For an odd prime put $T = p$ and take instead

$$(s_0, s_2, s_4, s_6) = (T^6, T^6 + T^7, T, T + T^2), \quad (t_0, t_2, t_6) = (T^5, T^{-2}, T),$$

with the same formula for t_4 replacing 4 by T^2 . The numerator is a unit. The output coefficient at exponent 6 is exactly p^2 , and each other nonzero coefficient has one contribution. For both recipes the complete height list, at exponents $2, \dots, 10$, is

$$17, 11, 7, 4, 2, 1, 1, 2, 5.$$

The slopes $-6, -4, -3, -2, -1, 0, 1, 3$ increase strictly. Thus $V = U = 8$. Apply the isometry (11) below with $a = s_0, b = s_2$, and clear a common denominator. Two coefficients become axis vectors and the remaining three have three coordinates, giving $S = 11$ and one integer-square norm.

For $p = 2$ this gives the explicit 26-bit integer example

$$\begin{aligned} P &= 519168x^2 + 121680x^3 - 37681x^4 + 9126x^6, \\ Q &= -66453504 - 3204240x^3 + 1169649x^4 - 1242150x^6, \\ R &= -624624x^3 + 209937x^4 - 106470x^6. \end{aligned}$$

The sole nonzero norm is 16224^2 , at exponent 3. Its output heights are the displayed list shifted upward by ten. Every unit edge has a simple linear residual factor, so each example has exactly eight simple nonzero $\mathbb{Q}_p$ roots.

For $n \leq 4$, there are at most $\binom{n}{2} + 1$ contributions when $k = 1$, so $V \leq \binom{n}{2} \leq 2n - 2$. Together with the five-support bound and the six-support construction in Section 7, this shows that six is the smallest vector-support size for which $V > 2n - 2$ is possible over $\mathbb{Q}_p$, for every prime p .

5 The finite necessary-condition certificate

5.1 Complete geometry and vertex patterns

The exceptional diagonal exponent is already a pair exponent. If $V \geq 12$, at least thirteen distinct pair sums are needed, among the fifteen pairs $i < j$. Equality of distinct pair sums can occur only as $A_i + A_l = A_j + A_k$ with $i < j < k < l$. There are fifteen possible elementary equalities. Every pair partition with at least thirteen groups is generated by at most two such equalities: choose a spanning tree within each group. Taking all equivalence closures gives 119 partitions. There are ten reflected primary placements, hence 1,190 geometry cases.

The geometry certificate accounts for all cases as follows.

Outcome	Cases	Exact evidence
Forced unlisted collision	915	Rational row-space identity between two pair sums declared distinct
Impossible strict order	48	Integer linear contradiction from the support equalities and $A_{i+1} > A_i$
Feasible geometry	227	Rational ordered support with exactly the specified pair-sum equalities

In a feasible partition with N output groups, mark thirteen strict vertices and leave $N - 13$ groups unmarked. The unmarked groups are allowed to be vertices too. This yields 1,050 marked cases with no pair collisions, 1,176 with one, and 133 with two: 2,359 in total. The classification uses no numerical bound on the exponents.

5.2 Convexity rules

Let ψ be the lower Newton boundary after normalization. Extend it outside the output interval with affine rays of slopes strictly below the first edge slope and strictly above the last. Choose the left slope nonpositive and the right slope nonnegative. Then all marked vertices, including endpoints, remain strict vertices of the extension, and ψ stays positive. Set $h_{ij} = \psi(A_i + A_j)$ for $0 \leq i \leq j \leq 5$.

At marked groups, $H_e = \psi(e)$; at unmarked groups, $H_e \geq \psi(e)$. Identically equal exponents have equal h values, and $h_{pq} > 0$. The following rules are immediate from convexity:

$$h_{ij} + h_{kl} \geq h_{il} + h_{kj} \quad (i < k, j < l), \quad (6)$$

$$h_{pr} + h_{qr} \geq 2h_{ar}. \quad (7)$$

The first is strict if either inner argument is marked; the second is strict if $A_a + A_r$ is marked. For the first, both inner arguments are strictly between the outer arguments and their sums agree.

We also use all equal-barycenter relations implied by the primary and collision equalities. If $u + v = s + t$ and $u < s, u < t$, convexity gives

$$\psi(u) + \psi(v) \geq \psi(s) + \psi(t), \quad (8)$$

strictly if either inner argument is marked. This is encoded as the finite disjunction

$$u \geq s \quad \text{or} \quad u \geq t \quad \text{or} \quad \psi(u) + \psi(v) \geq \psi(s) + \psi(t).$$

The support relations are checked by exact rational row reduction. The inequalities are not inferred from one sampled support realization.

Lemma 5.1 (Rigidity of two additive height equations). *Let ϕ be finite and convex on an interval containing $[0, a + b]$, where $s > 0$, $a > s$, and $b > 2s$. If*

$$\phi(a + s) - \phi(a) = \frac{1}{2}\phi(2s), \quad \phi(a + b) = \phi(a) + \phi(b),$$

then $\phi(t) = \lambda t$ on $[\min(a, 2s), a + b]$, where $\lambda = \phi(2s)/(2s)$.

Proof. Subtract λt and write the resulting convex function as χ . Then $\chi(2s) = 0$, $\chi(a + s) = \chi(a)$, and $\chi(a + b) = \chi(a) + \chi(b)$. If $a \leq b$, monotonicity of equal-length increments gives $\chi(b + s) \geq \chi(b)$, while convexity on the equal-barycenter pairs $(2s, a + b)$ and $(a + s, b + s)$ gives the reverse inequality. Equality forces affinity on $[2s, a + b]$. The equal values at $b, b + s$ force slope zero, hence value zero. The equality $\chi(a) = 0$ extends this interval to a if $a < 2s$. If $a > b > 2s$, comparison with the zero secant on $[a, a + s]$ gives $\chi(a) \leq 0$, and convexity gives $\chi(b) \leq 0$. Since $b > s$, convexity also gives $\chi(a + b) \geq \chi(a)$, hence $\chi(b) \geq 0$. Thus $\chi(b) = \chi(a) = 0$. Equality on the chord with endpoints $2s, a + b$ at the interior point b forces affinity and value zero throughout that interval. $\square$

5.3 Coefficient and Gram rules

For a coefficient group not containing the primary, the minimum among its pair-entry heights and its output height occurs at least twice. In particular, a singleton group has $H = w$. The primary group already has two zero-height contributions, its pair and the exceptional diagonal; we impose no further coefficient rule there.

For four ordered indices $i < j < k < l$, put $X = C_{ij}C_{kl}$, $Y = C_{ik}C_{jl}$, and $Z = C_{il}C_{jk}$. Their Gram determinant is

$$X^2 + Y^2 + Z^2 - 2XY - 2XZ - 2YZ = 0 \quad (9)$$

if all four vectors are null. If the exceptional vector is present, add $4d_a C_{rs} C_{rt} C_{st}$, where r, s, t are the other indices. Thus the minimum among

$$2(w_{ij} + w_{kl}), \quad 2(w_{ik} + w_{jl}), \quad 2(w_{il} + w_{jk}) \quad (10)$$

and, in the exceptional case,

$$2\epsilon + w_{rs} + w_{rt} + w_{st}$$

must occur at least twice. A unique minimum in this list would remain unique in the full determinant, because a mixed term has height ϵ plus the average of two square-term heights and $\epsilon \geq 0$. The argument remains valid in residue characteristic two.

For an exceptional quartet whose six pairs are singleton marked groups and avoid the primary, strict convexity makes the nested matching uniquely smallest among the three matchings. The additional diagonal term must tie its square exactly. This forced linear equality is the certificate rule named `clean_gram`.

5.4 Direct obstructions and finite disjunction trees

Let L contain the primary and all pairs in nonsingleton groups, and let U_0 contain all pairs in nonsingleton or unmarked groups. A null quartet $i < j < k < l$ is impossible whenever

$$\{ij, kl, ik, jl\} \cap L = \emptyset, \quad \{il, jk\} \cap U_0 = \emptyset.$$

Indeed, the nonnested entries are at least their boundary heights, while the nested entries are at most their boundary heights: they equal the marked singleton heights, or are the primary at zero below its positive boundary. The nested arguments are marked strict vertices. Strict convexity therefore makes the nested matching uniquely smallest, contrary to (9). This excludes 1,397 marked cases.

Each of the remaining 962 cases is encoded by the preceding linear rules and finite disjunctions. A repeated minimum is encoded by choosing every possible pair of equal minimal entries. At an internal proof-tree node, every branch of the selected disjunction is present. A leaf gives integer multipliers for its accumulated linear constraints. Multipliers of inequalities are nonnegative, multipliers of equalities are unrestricted, and some strict inequality has positive weight. The weighted sum of the linear forms is identically zero, giving $0 > 0$.

The certificate has 9,085 nodes and 6,627 such leaves. Exactly six cases require the additional equal-barycenter rules. The solver-free checker independently enumerates the partitions, verifies all geometry certificates, regenerates the marked patterns, checks all direct null obstructions, audits each rule instance, verifies every branch, and evaluates each integer identity exactly. All 2,359 cases are excluded. This proves the all-edge assertion of Theorem 1.1.

6 The unit-edge bound

Suppose $U \geq 11$. The established $V \leq 11$ gives $V = U = 11$, so the twelve strict vertices have consecutive integer exponents and the output span is eleven. Apply Lemma 2.1. All old vertices persist, and the bound $V \leq 11$ permits no additional vertex. The output interval is therefore unchanged. Since all pair entries are now finite and the hidden diagonal exponent is a pair sum,

$$A_4 + A_5 - A_0 - A_1 = 11.$$

Translate A_0 to zero. The integer support order gives $A_4 - A_1 \geq 3$, whence $A_5 \leq 8$. This bound follows from the assumed eleven unit edges; it is not an imposed search limit.

Exactly four six-element subsets of $\{0, \dots, 8\}$ containing zero have twelve consecutive pair sums:

$$\begin{aligned} \{0, 1, 2, 3, 4, 8\}, & \quad \{0, 2, 3, 4, 5, 8\}, \\ \{0, 3, 4, 5, 6, 8\}, & \quad \{0, 4, 5, 6, 7, 8\}. \end{aligned}$$

After reflection $a \in \{1, 2\}$, and there are ten possible primary placements across these supports. Mark every one of the twelve output groups. The same linear rules exclude all ten cases, with 749 nodes and 567 exact contradiction leaves; two cases use equal-barycenter convexity. The separate checker `verify_unit_six.py` regenerates the supports and primary placements and verifies the full certificate. This proves $U \leq 10$.

7 Simultaneous sharpness and a compact formula

Set $c_3 = (0, 2, 1)$ and $c_i = t_i(-s_i^2, 1, s_i)$ for $i \in \{0, 2, 4, 6, 8\}$. Over $\mathbb{Q}_p$ put $T = p$ and use the parameters in Table 1. Define

Case	A	B	D	E	u	v	w	H_6	H_{10}
$p = 2$	39	41	18	13	10	-17	-2	20	18
p odd	33	34	14	11	9	-14	-1	16	14

Table 1: Parameters for the sharp six-support formula.

$$\begin{aligned}
s_0 &= T^A, & s_2 &= T^A + T^B, & s_4 &= T^{-1}, & s_6 &= T^D, & s_8 &= T^E, \\
t_0 &= T^u, & t_2 &= T^v, & t_6 &= T^w, \\
t_4 &= \frac{1 - T^{H_6} - t_0 t_6 (s_0 - s_6)^2}{t_2 (s_2 - s_4)^2}, & t_8 &= \frac{-T^{H_{10}} - t_4 t_6 (s_4 - s_6)^2}{t_2 (s_2 - s_8)^2}.
\end{aligned}$$

The directions are distinct and all denominators are nonzero. The coefficient norms are zero except $\mathfrak{q}(c_3) = 1$.

The definitions force the output coefficients at 6 and 10 to be T^{H_6} and $T^{H_{10}}$. The numerator defining t_4 is a unit: $H_6 > 0$ and $u + w + 2D > 0$. Hence $\nu_p(t_4) = 2 - v$. The two terms in the numerator defining t_8 have unequal valuations, with $w - v < H_{10}$. Therefore $\nu_p(t_8) = w - 2v - 2E$.

With $\epsilon = \nu_p(2)$, the output heights in exponent order 2, 3, ..., 12, 14 are

$$\begin{aligned}
&u + v + 2B, & \epsilon + u + A, & u - v, & \epsilon + v + A, & H_6, \\
&\epsilon - v, & v + w + 2D, & \epsilon + w + D, & H_{10}, \\
&\epsilon + w - 2v - E, & w - 3v - 2E, & 2w - 2v.
\end{aligned}$$

Every unforced coefficient has a unique minimum contribution. At exponent 8 the two heights are $u + w - 2v$ and $v + w + 2D$, and the latter is smaller because $u - 3v > 2D$. All other unforced coefficients have a single pair. The resulting tables are

Exponent	2	3	4	5	6	7	8	9	10	11	12	14
$p = 2$	75	50	27	23	20	18	17	17	18	20	23	30
p odd	63	42	23	19	16	14	13	13	14	16	19	26

For odd p the consecutive slopes are

$$-21, -19, -4, -3, -2, -1, 0, 1, 2, 3, 7/2;$$

for $p = 2$ the first two become $-25, -23$. They are strictly increasing. Thus all twelve points are strict vertices, the first ten edges have width one, and the last has width two. This proves $(V, U) = (11, 10)$.

For integer coordinates, put $a = s_0$, $b = s_2$, and apply

$$(P, Q, R) \mapsto \frac{(P + 2aR - a^2Q, -P - 2bR + b^2Q, -P - (a + b)R + abQ)}{a - b}. \quad (11)$$

Direct expansion verifies preservation of $PQ + R^2$. The vectors at 0 and 2 become coordinate-axis vectors; the other four have three nonzero coordinates for the displayed rational parameters. Clearing a common denominator M gives $S = 1 + 1 + 4 \cdot 3 = 14$, a sole nonzero norm M^2 , and a common output-height shift $2\nu_p(M)$. The dyadic frozen witness has maximum input length 228 bits.

If $\nu(2) = 0$ in a general nontrivially valued field, choose any T with positive valuation and use the odd row of the table. The displayed heights are multiplied by $\nu(T)$. The valuation proof uses

only unit leading coefficients. Independently, the exact formal checker clears denominator factors with constant term one and expands in $\mathbb{Z}[T, T^{-1}]$; every leading coefficient is a signed power of two. This verifies specialization in every odd positive characteristic too.

If $\nu(2) > 0$, then K has characteristic zero and $\nu|_{\mathbb{Q}}$ is $\nu(2)$ times the ordinary dyadic valuation. Indeed, every odd integer is a unit by the ultrametric inequality. The rational dyadic formula therefore specializes to K . This proves sharpness in Theorem 1.1. A trivial valuation has at most one strict lower edge, explaining the nontriviality qualification.

Finally, over $\mathbb{Q}_p$ every unit edge has a linear residual polynomial with a nonzero simple root. Hensel's lemma gives one simple root of its valuation. The last edge has nonintegral slope $7/2$ and gives no $\mathbb{Q}_p$ root. This proves Corollary 1.2.

8 An all-size one-norm cancellation family

We prove the following uniform construction directly.

Theorem 8.1. *For every prime p and integer $n \geq 8$, there are integer polynomials P, Q, R with*

$$k = 1, \quad V_p = 2n + 1, \quad U_p = 2n, \quad S = 3n - 4.$$

Their vector support is $\{0, 1, 2, 4\} \cup \{7, \dots, n + 2\}$. For each p , one fixed nonzero integer-square norm works for every n . The input coefficient bit lengths are $O(n^2 \log p)$, and the output has exactly $2n$ simple nonzero $\mathbb{Q}_p$ roots, of distinct valuations.

8.1 The eight-support core

Set $T = p$, $c_4 = (0, 2, 1)$, and $c_i = t_i(-s_i^2, 1, s_i)$ elsewhere. In both residue characteristics put

$$s_0 = T^{-3}, \quad s_1 = T^{-4}, \quad s_2 = 1 + T^{35}, \quad s_9 = 1 + T^{152}, \quad s_{10} = 1 + T^{144}.$$

For $p = 2$ take

$$\begin{aligned} s_7 &= 1 + T^{141}, & s_8 &= 1 + (T^{18} - 1)T^{142}, \\ t_0 &= T^{226}, & t_1 &= \frac{T^{153}}{(1 - T^4)^2}, & t_2 &= T^{74}. \end{aligned}$$

For odd p take

$$\begin{aligned} s_7 &= 1 + (T - 1)T^{141}/2, & s_8 &= 1 + (T^{18} - 1)T^{142}/2, \\ t_0 &= T^{225}, & t_1 &= \frac{T^{152}}{(1 - T^4)^2}, & t_2 &= T^{73}. \end{aligned}$$

Write $D_{ij} = (s_i - s_j)^2$ and define t_7, t_8, t_9, t_{10} by

$$\begin{pmatrix} t_1 D_{17} & t_0 D_{08} & 0 & 0 \\ t_2 D_{27} & t_1 D_{18} & t_0 D_{09} & 0 \\ 0 & t_2 D_{28} & t_1 D_{19} & t_0 D_{0,10} \\ -2s_7(1 - s_7) & 0 & t_2 D_{29} & t_1 D_{1,10} \end{pmatrix} \begin{pmatrix} t_7 \\ t_8 \\ t_9 \\ t_{10} \end{pmatrix} = \begin{pmatrix} 1 - T^{61} \\ -T^{48} \\ -T^{36} \\ -T^{25} \end{pmatrix}. \quad (12)$$

The null-null cross coefficients are $-t_i t_j D_{ij}$, and those with c_4 are $2t_i s_i(1-s_i)$. Thus these equations prescribe the output coefficients at 8, 9, 10, 11 exactly. The dyadic valuation matrix is

$$\begin{pmatrix} 145 & 220 & \infty & \infty \\ 144 & 145 & 220 & \infty \\ \infty & 144 & 145 & 220 \\ 142 & \infty & 144 & 145 \end{pmatrix};$$

for odd p , subtract one from every finite entry. The diagonal determinant term is uniquely minimal, of valuation 580 or 576: every nontrivial permutation cycle uses an upper entry with excess 75, whereas the total decreases from the other entries in that cycle are at most five. Put $K = 145$ or 144, respectively, and normalize

$$a = T^K t_7, \quad b = T^{K+1} t_8, \quad c = T^{K+2} t_9, \quad d = T^{K+2} t_{10}.$$

Multiplying the rows by $1, T, T^2, T^3$, respectively, the first three normalized rows give

$$a = 1 - T^{61} \pmod{T^{74}}, \quad b = -a - T^{49} \pmod{T^{74}}, \quad c = -b - T^{38} \pmod{T^{75}}.$$

The normalized $t_1 D_{1j}$ correction is $((1 - T^4 s_j)/(1 - T^4))^2$, which is one to precision greater than 140. The normalized D_{2j} correction is one to precision at least 106. These exceed all precisions used in the elimination. The fourth row gives

$$a + c + 2d = -2^{28} \pmod{2^{49}} \quad (p = 2), \quad (T - 1)a + c + Td = -T^{28} \pmod{T^{49}} \quad (p \neq 2).$$

The normalized matrix is integral with determinant valuation one. Its cofactor formula first bounds all four unknown valuations below by -1 . The first three rows then give their congruences with precision reduced by one; the fourth gives $Td = -T \pmod{T^{28}}$. This proves integrality and recovers the full displayed precisions. Thus $a, c \equiv 1$ and $b, d \equiv -1 \pmod{T^{27}}$. The coefficient at exponent 12 is

$$-T^{-3} \left((T^{18} - 1)s_8 b + \left(\frac{s_2 - s_{10}}{T^{35}} \right)^2 d \right),$$

whose bracket is $-T^{18}$ modulo T^{27} . The four solved coefficients and all remaining unique minima now give the strict dyadic vertices

$$\begin{aligned} &(1, 371), (2, 294), (3, 219), (5, 146), (6, 110), (7, 75), \\ &(8, 61), (9, 48), (10, 36), (11, 25), (12, 15), (13, 6), (14, -2), \\ &(15, -9), (16, -10), (17, -10), (18, -9), (19, -6). \end{aligned}$$

For odd p , replace these boundary heights as follows:

e	1	2	3	5	6
$\nu(f_e)$	369	292	217	144	108
e	15	16	17	18	19
$\nu(f_e)$	-7	-8	-8	-7	-4

The sole additional nonzero point has exponent 4 and height 221 or 219, respectively, strictly above the boundary. The dyadic slopes are

$$\begin{aligned} &-77, -75, -73/2, -36, -35, -14, -13, -12, -11, \\ &-10, -9, -8, -7, -1, 0, 1, 3. \end{aligned}$$

At odd primes, -35 becomes -33 , and -7 becomes -5 . In either case there are seventeen strict edges and sixteen unit edges. The sole nonunit edge, from exponent 3 to 5, has slope $-73/2$.

8.2 The tail and its complete height comparison

For $11 \leq j \leq n+2$ append the dyadic tail

$$t_j = T^{4j^2-66j+113}, \quad s_j = 1 + T^{144} + \sum_{r=11}^j T^{6r+89},$$

or the odd-prime tail

$$t_j = T^{4j^2-66j+114}, \quad s_j = 1 + T^{144} + \sum_{r=11}^j T^{6r+88}.$$

Put $h_j = \nu_p(t_j)$, using the displayed quadratic also at $j = 10$. For $10 \leq i < j$, the direction distance has valuation $6i + 95$ in the dyadic case and $6i + 94$ otherwise. In both cases the cross height is

$$4i^2 + 4j^2 - 54i - 66j + 416.$$

At a fixed sum $e = i + j \geq 21$, this is a strictly convex quadratic in i , with real minimizer $e/2 - 3/4$. The unique allowed integer minimizer is the adjacent pair for odd e and the gap-two pair for even e . Its height is

$$H_e = 2e^2 - 60e + 412. \quad (13)$$

The pair $(9, 11)$ has height 12 for $p = 2$ and 14 otherwise, at exponent 20. More generally, the height of $(9, j)$ minus H_{j+9} is $2(j-10)(j-11)$ for $p = 2$, and that expression plus two otherwise. It is positive for every $j \geq 12$.

For the other old indices, the new cross heights are $h_j + C_i$, where

i	0	1	2	4	7	8
$C_i (p = 2)$	220	145	144	145	137	138
$C_i (p \neq 2)$	219	144	143	144	138	139

For $p = 2$, subtracting H_{j+i} gives

i	difference	range for $j + i \geq 20$
0	$2j^2 - 6j - 79$	$j \geq 20$
1	$2j^2 - 10j - 96$	$j \geq 19$
2	$2j^2 - 14j - 43$	$j \geq 18$
4	$2j^2 - 22j + 54$	$j \geq 16$
7	$2j^2 - 34j + 160$	$j \geq 13$
8	$2j^2 - 38j + 191$	$j \geq 12$

For odd p , the first four differences are unchanged and the last two increase by two. Each polynomial is positive and increasing in the indicated range. For the finitely many $j \geq 11$ with $j + i < 20$, substitution in the core table also gives strict inequalities; the smallest margin is one, at $(i, j) = (1, 11)$. At exponent 20, every competitor to $(9, 11)$ is strictly higher, including in the odd case where the boundary height is 14 instead of $H_{20} = 12$.

Hence no core vertex is lost and all new coefficients have a unique minimum contribution. The added vertices are $(20, 12)$ for $p = 2$, $(20, 14)$ for odd p , and (e, H_e) for $21 \leq e \leq 2n + 3$. The first new slopes are $18, 22, 26, 30, \dots$ for $p = 2$ and $18, 20, 26, 30, \dots$ for odd p . They increase strictly, starting above the core's final slope three. Each appended coefficient therefore adds two unit edges. This proves $V = 17 + 2(n - 8) = 2n + 1$ and $U = 16 + 2(n - 8) = 2n$ for all $n \geq 8$.

8.3 Integer coefficients, sparsity, and roots

For the integer realization, put $a = s_0, b = s_1$ and apply

$$(p_0, q_0, r_0) \mapsto \frac{(p_0 + 2ar_0 - a^2q_0, -p_0 - 2br_0 + b^2q_0, -p_0 - (a + b)r_0 + abq_0)}{a - b}. \quad (14)$$

Direct expansion proves that this is a $\mathfrak{q}$ -isometry. The first two coefficient vectors become coordinate axes, and all others have three nonzero coordinates. Hence $S = 3n - 4$. Each matrix denominator divides $p^4(p - 1)$ and every new scale denominator divides p^{129} . The core denominator already contains p^{K+6} , from its exponent-9 coefficient, and contains $p - 1$, from the exponent-2 coordinate $p^{\nu(t_2)-4}(p^4 + p^{39} - 1)^2/(p - 1)$. Its numerator is coprime to $p - 1$. Thus the same core denominator clears every tail, giving one fixed integer-square norm. The displayed exponents give $O(n^2 \log p)$ input bits. This completes Theorem 8.1.

Each unit edge gives one simple nonzero $\mathbb{Q}_p$ root, by its linear residual polynomial. The remaining slope $-73/2$ is nonintegral at every prime. Consequently the nonzero $\mathbb{Q}_p$ root count is exactly $2n$.

In particular, the candidate inequalities $V \leq 2n + 2k - 4$ and $U \leq 2n + 2k - 4$ fail already at $k = 1$. The exceptional coefficient in this construction is interior, at exponent four. The family does not represent a prescribed target polynomial with an exceptional constant term.

9 Computational verification

The ancillary archive `verification.zip` contains the exact certificates, example coefficients, and checking programs. After extracting the archive, `python -B -X utf8 verify.py` checks the file manifest and runs all five verification programs. The programs can also be run individually:

```
python -B -X utf8 verify_sharp_six.py
python -B -X utf8 verify_unit_six.py
python -B -X utf8 verify_unit10_examples.py
python -B -X utf8 verify_positive_control.py
python -B -X utf8 verify_families.py
```

They require no solver. The first two regenerate the rule systems, independently audit the formula and justification metadata of every instance, and check all finite coverage and integer contradictions. The third imports no constructor: it expands the frozen integer inputs by ordered scalar convolution, checks every coefficient against every proposed supporting line, checks all coefficient norms, and lifts sixty roots modulo p^{24} at $p = 2, 3, 5, 7, 11, 17$. It also verifies the formal field-uniform sharpness calculation. The fourth evaluates 192 axioms and 4,176 disjunctions on a genuine eleven-edge equality example and rejects invalid arithmetic-certificate controls.

The fifth checker verifies the results in Sections 4 and 8: six frozen five-support examples, twenty-five family members through $n = 35$ at six primes, and 144 simple root lifts to precision p^{24} . It checks the twelve polynomial tail inequalities on their full integer ranges using their coefficients and monotonicity, the sixty-four finite core comparisons, and uniqueness of the core determinant minimum in both residue-characteristic cases. The general upper bound at five supports and the uniform family assertions follow from the written proofs.

Z3 was used to generate the certificates. The checking programs verify their integer identities directly and require only the Python standard library. Each selected proof file is hashed in a manifest; the checkers validate these hashes and record their own source hashes. Python optimization

flags are rejected because the implementation uses assertions for exact checks. The archive is self-contained. The mathematical reduction to the verified finite systems is proved in the preceding sections.

10 Further questions

The convexity argument uses additive relations among exponent sums before introducing their boundary heights. It resolves six patterns that the principal Gram conditions and the basic Monge and primary-midpoint inequalities did not exclude. The resulting finite proof depends on at most two pair-sum losses when thirteen vertices are demanded from fifteen pairs. This counting feature is specific to the present extremal problem. No extrapolation to arbitrary support size follows from the certificate.

The next broader questions concern a uniform bound for growing n with one nonzero norm, and restrictions on endpoint-normalized representations of specified completely convex outputs. Theorem 8.1 rules out the candidate bound $2n + 2k - 4$ for both edge statistics. The sharp five- and six-support results and this all-size construction leave open a uniform upper bound for growing support. None of these results supplies an unrestricted circuit separation.

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References

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